

20CYS443 Biometrics and Security L-T-P-C:3-0-0-3

Pre-Requisite: 20CYS212 Multimedia Processing Course

Objectives

- Understand the technological uplifts with biometrics compared to traditional securing mechanisms and standards applied to security
- To understand the concepts of different types of biometrics and to enable design of biometric system and its privacy risks
- To familiarize with biometric interface and biometric applications

Course Outcome

CO1: Apply biometric matching to identify algorithms for finger biometric technology, check the performance measures and its security

CO2: Develop facial biometric, iris biometric, voice biometric, physiological biometrics for identification technology.

CO3: Understand different types of user interfaces.

CO4: Designing privacy sympathetic biometric systems and identifying the area of biometric applications

CO-PO Mapping

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO														
CO1	2	3	3	2	2	3		3					3	3
CO2	2	3	3	2	2	3		3					3	3
CO3		3	3	1	2	3		2					2	2
CO4	2	3	3	3	2	3		3					3	3

Syllabus

Biometric Fundamentals and Standards: Biometrics versus traditional techniques, Characteristics, Key biometric processes: Verification -Identification -Biometric matching, Performance measures in biometric systems, Assessing the privacy risks of biometrics - Designing privacy sympathetic biometric systems, Different biometric standards, Application properties.

Physiological Biometrics: Facial scan, Ear scan, Retina scan, Iris scan, Finger scan, Automated fingerprint identification system, Palm print, Hand vascular geometry analysis, Knuckle, DNA, Dental, Cognitive Biometrics -ECG, EEG.

Behavioral Biometrics: Signature scan, Keystroke scan, Voice scan, Gait recognition, Gesture recognition, Video face, Mapping the body technology.

User interfaces: Biometric interfaces: Human machine interface -BHMI structure, Human side interface: Iris image interface -Hand geometry and fingerprint sensor, Machine side interface: Parallel port -Serial port -Network topologies, Case study: Palm Scanner interface.

Biometric applications: Categorizing biometric applications, Application areas: Criminal and citizen identification –Surveillance -PC/network access -E-commerce and retail/ATM, Costs to deploy, Issues in deployment, Biometrics in medicine, cancellable biometrics.

Text Books/References:

1. Anil K Jain, Patrick Flynn and Arun A Ross, Handbook of Biometrics, Springer, US; 2010
2. John R Vacca, Biometric Technologies and Verification Systems, Elsevier, USA; 2009
3. Samir Nanavati, Michael Thieme and Raj Nanavati, Biometrics –Identity Verification in a Networked World, John Wiley and Sons ; 2003
4. Paul Reid, Biometrics for Network Security, Pearson Education; 2004
5. ReidM. Bolle et al, Guide to Biometrics, Springer, USA; 2004
6. David D Zhang, Automated Biometrics: Technologies and Systems, Kluwer Academic Publishers; 2000.

Table 2A: Internal Assessment Components

Component	TL Courses (4 & 3 Credits)	Theory Courses (3 Credits)	Lab Courses (1 Credit)
CA-1	8	7	—
CA-2 (Theory) / Lab Evaluation-1 (TL) (including viva voce)	7	8	20
Mid-Term Examination	30 (20 Theory + 10 Lab)	30	20
CA-3	8	7	—
CA-4 (Theory) / Lab Evaluation-2 (TL) (including viva voce)	7	8	20
Total Internal Assessment Marks	60	60	60

Aug 17 midterm

Oct 16 – last working day

Oct 21- end Sem exam starts